

Video Editing Evaluation Using Auto-Supervised Learning

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The background features a light gray surface with several overlapping squares of varying shades of gray. Some squares have rounded corners, while others are sharp. They are arranged in a way that creates a sense of depth and layering, with some squares appearing to float above others.

01

Introduction

Main Points

Cinematography and Computer Science.

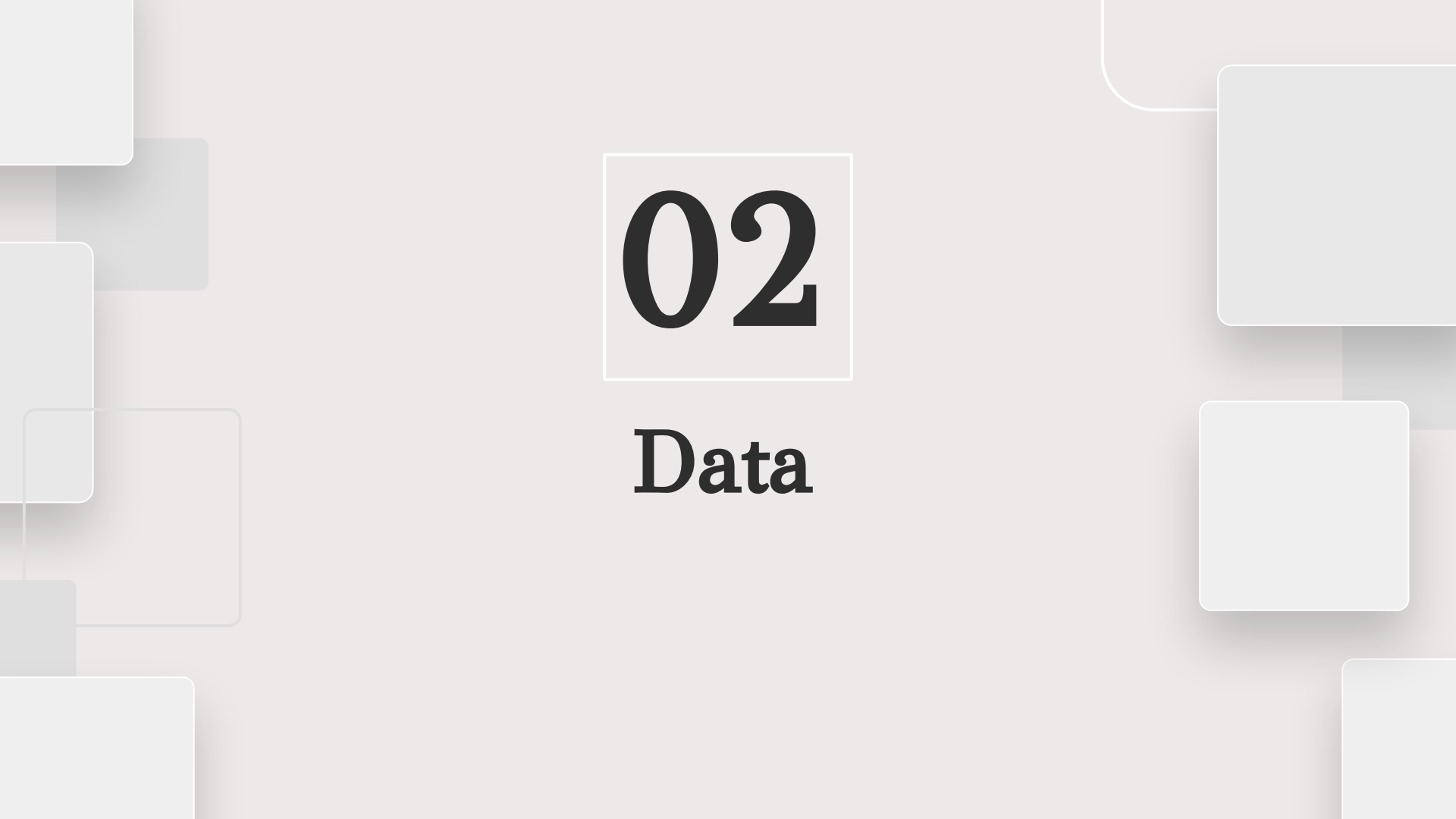
It is inevitable that at some point, computer science - and machine learning specifically - would be used in the process of filmmaking.

Why choose such a project?

We chose this project out of a mutual love for cinema and film.

How was Computer Science previously implemented in Cinematography?

Top uses of CS in Cinematography: Editing, Special Effects, Animations, Feature Extraction.

The background features several light gray squares of various sizes and orientations, some overlapping, creating a layered, architectural feel. The central text is framed by a thin white border.

02

Data

Initial Data Size

375,000 Cuts

60,000 scenes

Py-SceneDetect: OpenCV and FFMPEG based

What is a cut?

The background of the slide features several overlapping geometric shapes. On the left, there is a large light gray square. In front of it, towards the bottom, is a smaller white square with a thin gray border. To the right of the white square is a medium-sized gray square. Further up and to the right is a small gray square. A thin white line curves around the top right corner of the main title area.

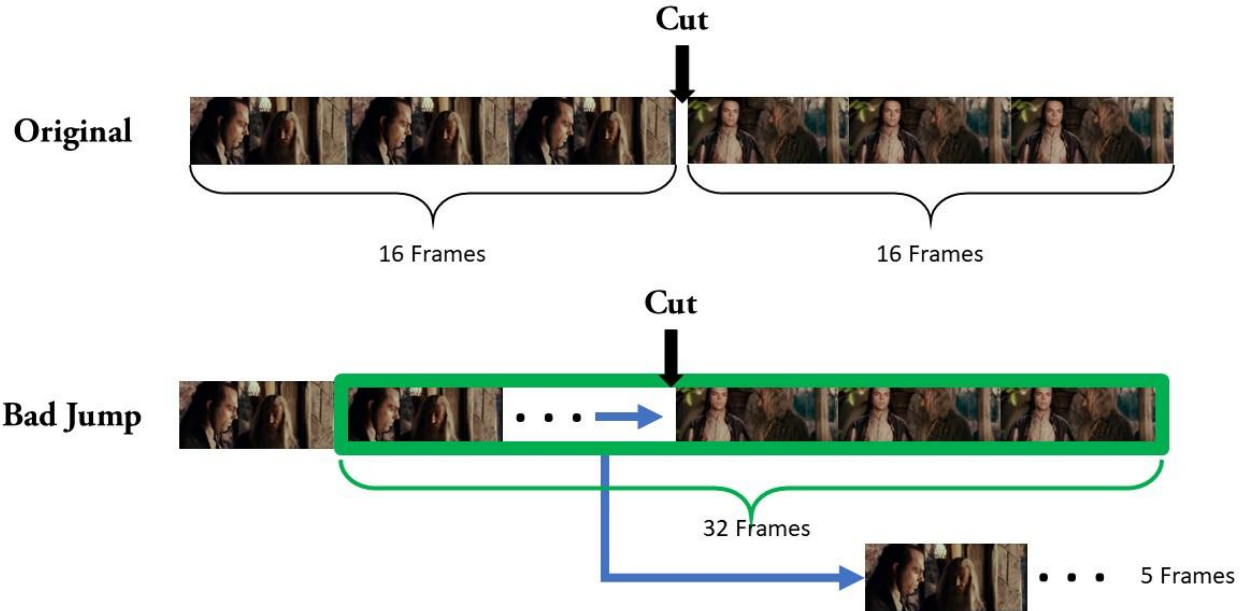
“In the post-production process of film editing and video editing, a cut is an abrupt, but usually trivial film transition from one sequence to another.”

- **Wikipedia**

There are many types of cuts



Creating the Bad Cut



Creating the Bad Cut

Original:

Cut



16 Frames

16 Frames

Bad Mix:

Original Cut

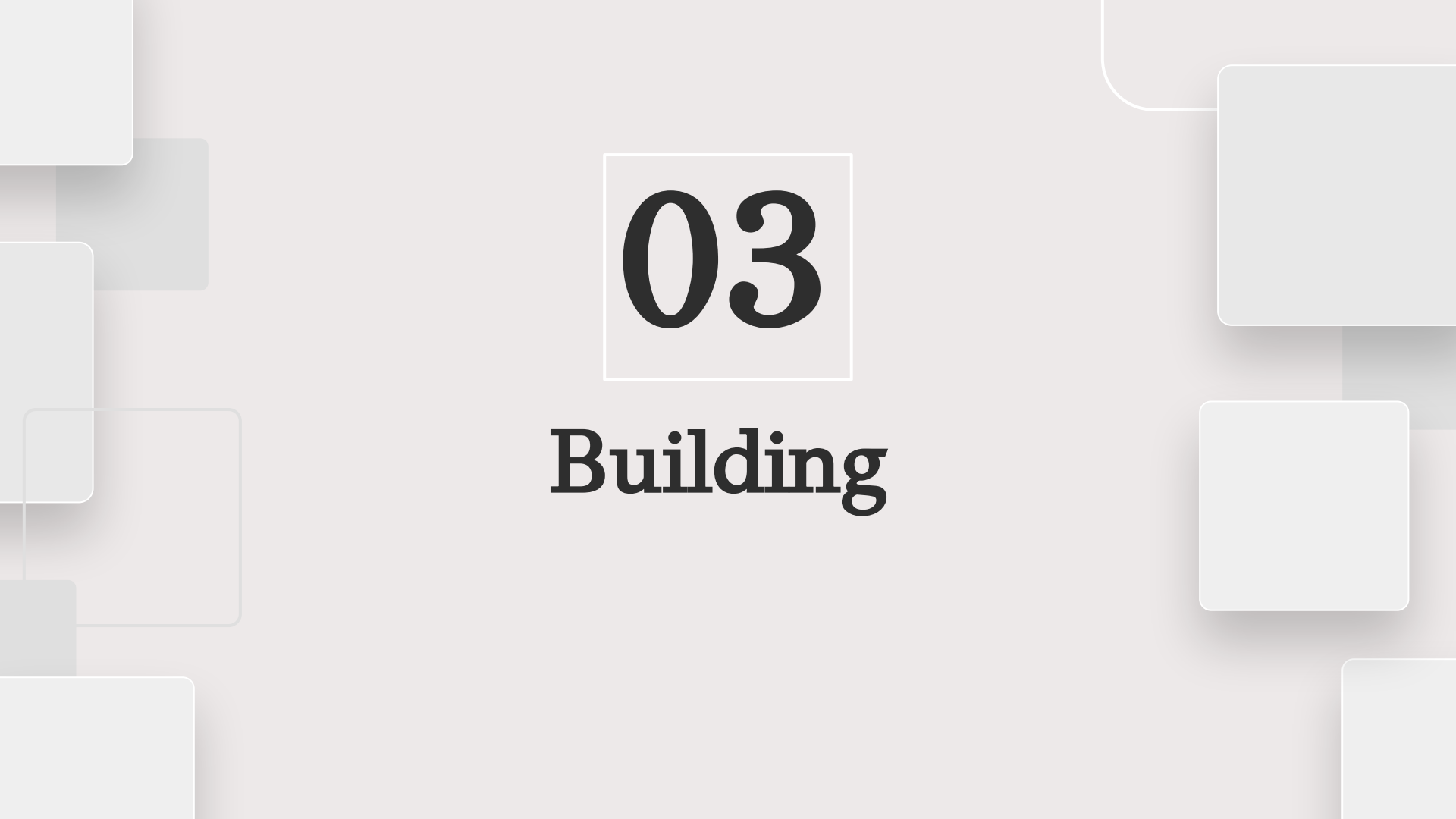


Different Shot



16 Frames

16 Frames

The background features several light gray squares of various sizes and opacities, some with rounded corners, arranged in a layered, geometric pattern on the left and right sides of the page.

03

Building

Setup

Model: **ResNet** (short for Residual Network)

Deep learning architecture introduced in 2015 by Microsoft Research. It's a specialized **Convolutional Neural Network (CNN)** designed to make the training of very deep networks practical and stable.

*Choice inspired by encouraging results of **MovieCut** project, research focused on the recognition of different types of cuts*

Setup

ResNet

Adam optimizer

— A Method for Stochastic Optimization

18 layers

— Layers that have trainable weights

R(2+1)D

— Pre-trained Weights

Kinetics-400

— Dataset that contains 400 human action classes

Cross Entropy Loss

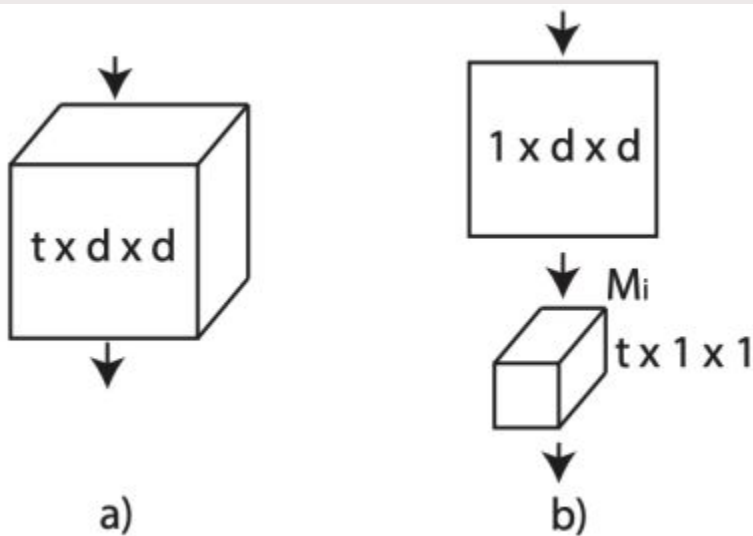
— Loss function used primarily in classification tasks

Batch size 4

— How many videos model can read at the same time

Setup

Carried out using a filter of size $t \times d \times d$ where t denotes the temporal extent and d is the spatial width and height



Splits the computation into a spatial 2D convolution followed by a temporal 1D convolution.

Low error rate and lighter

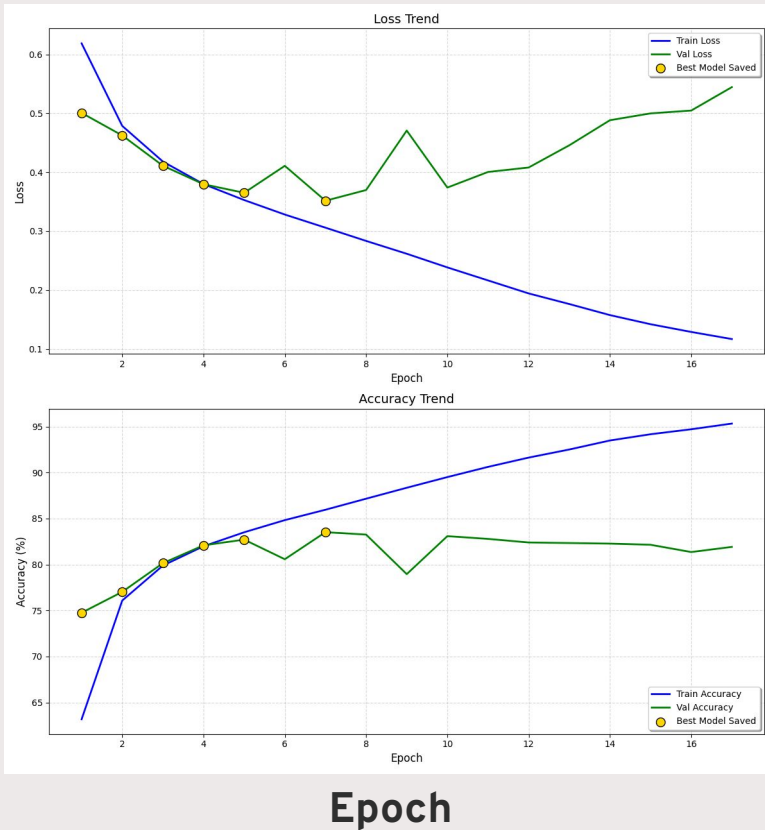
3D convolution vs (2+1)D

The background features a light gray surface with several overlapping squares of varying shades of gray and white. Some squares have rounded corners, while others are sharp. They are arranged in a way that creates a sense of depth and layering, with some squares appearing to float above others.

04

Experiments

Experiment with setup configuration



Loss Trend

Accuracy Trend

Training
Valuation
Model saved

Best epoch

Valuation loss : **0.3516**

Training loss : **0.3059**

Valuation accuracy : **83.52%**

Training accuracy : **85.96%**

Problem in real life

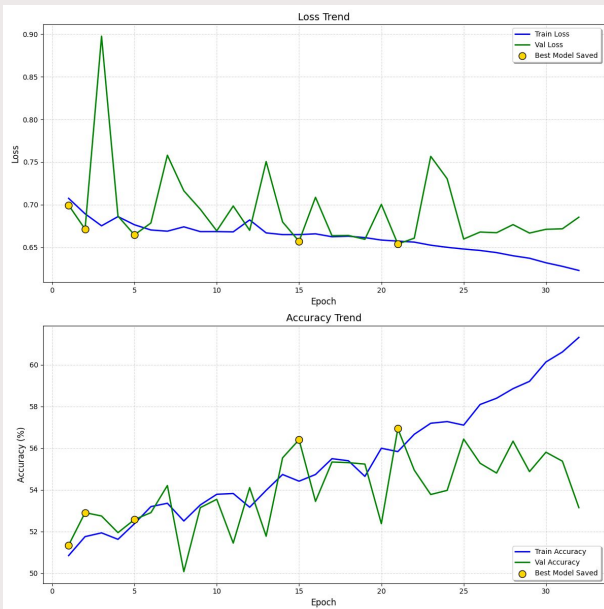
Model has **only 53%** accuracy on new dataset with 15,000 cuts, biased in positive cuts, so we continue the training on the new dataset with these techniques :

- **Learning Rate Scheduler**
 - **L2 Regularization**
 - **Cost-Sensitive Learning**
- New techniques added in the next trainings
- **Dropout**
 - **Data Augmentation**
 - **Negative sampling**
- Techniques tested one at a time

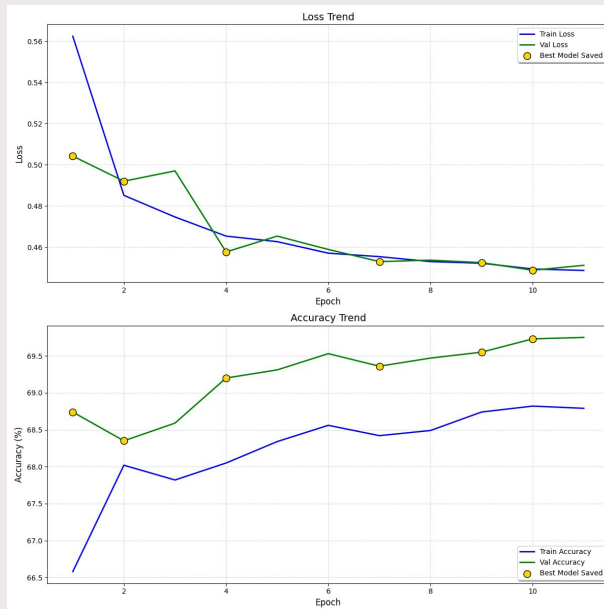
Experiments

■ Training
■ Valuation
● Model saved

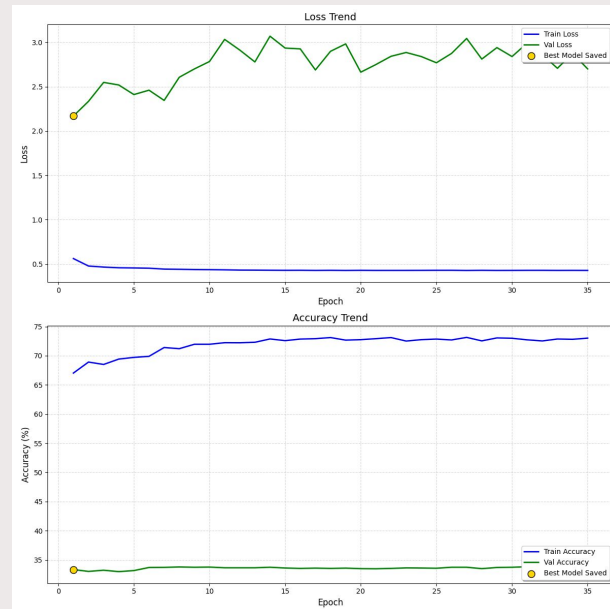
Data Augmentation
Best accuracy: **56.94%**



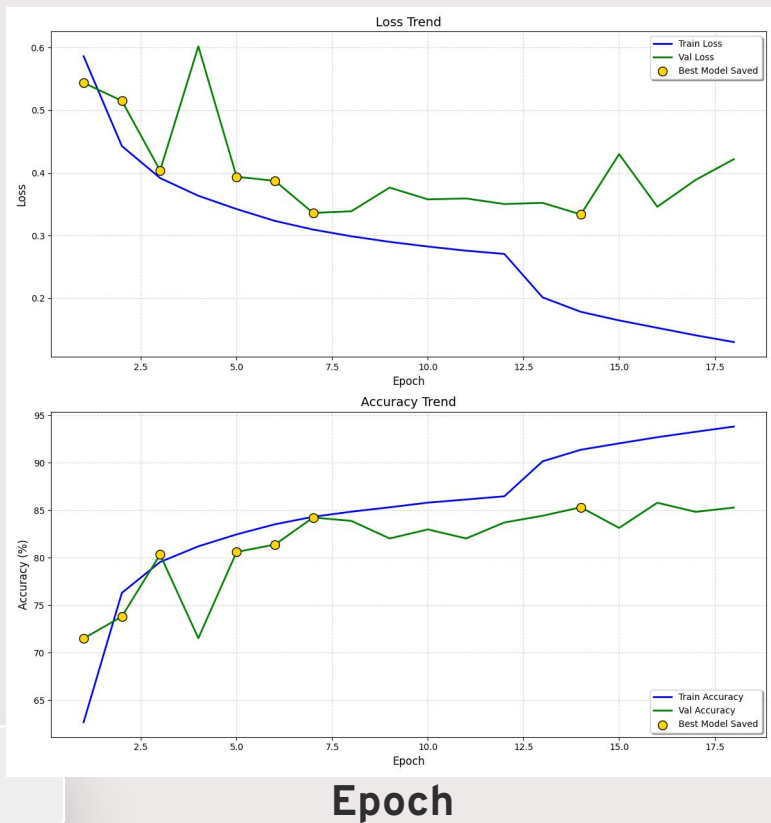
Negative sampling
Best accuracy: **69.73%**



Dropout
Best accuracy: **33.33%**



Final experiment



— Training
— Valuation
● Model saved

Loss Trend

Accuracy Trend

Merged 2 database with:
negative sampling
L2 Regularization
Cost-Sensitive Learning

Best epoch

Valuation loss : 0.3336
Training loss : 0.1780
Valuation accuracy : 85.96%
Training accuracy : 91.37%

Problem in real life

The model **gained 3%** in accuracy on the real-world dataset, **reaching 56%**, but with a higher cost in negative cuts it was completely biased on negative cuts.

Hypothesis : **Distribution Shifts problem**

when the data distribution of an algorithm's training dataset differs from the distribution it encounters when deployed.

The model learned the compression flaws of the training dataset rather than the cuts themselves.

The background features a light gray surface with several overlapping squares of varying shades of gray. Some squares have rounded corners and thin white outlines, while others are solid. They are arranged in a way that creates a sense of depth and geometric pattern, primarily along the left and right edges of the frame.

05

Recommendatio n

Recommendations for the future

The results are not quite satisfactory, but we looked for possible future avenues:

SlowFast Model

video recognition architecture developed by Meta AI (FAIR) in 2019

- **Slow Pathway:** Operates at a low frame rate and captures spatial semantics.
- **Fast Pathway:** Operates at a high frame rate with fewer channels, focusing on motion and temporal dynamics.

Recreate a new dataset from scratch

To make sure you have full control over all your files it is best to create a large database from scratch and to consider audio files of the scenes.

SlowFast model warnings

SlowFast can be very effective because:

- explicitly divides spatial and temporal features across two separate streams
- allows the network to learn both static environments and dynamic motion concurrently but independently.

BUT

- It remains a heavy model
- Managing both channels at the same time can be complex, each video has two distinct tensors

Conclusion

- Cinematography is a very interesting field for computer science application, but can benefit from an expert input.
- Audio and video, when it comes to filmmaking, cannot be separated.
- Perhaps a smaller frame count should be used to identify each cut, maybe 32 hide some useless details.

Thanks

Comments and questions in French
are welcome

By Riccardo VENTURINI & Halim NJEIM

*Thanks to the professor **Alexandre BRUCKERT** for following us in this research*